

Convex Optimization

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Define - Graph, Epigraph, Convex Set.

Convex Function.

Graph

The graph of a function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$, is the set of points in $\Omega \times \mathbb{R} \subset \mathbb{R}^{n+1}$ given by $\{[x, f(x)]^T : x \in \Omega\}$

Epigraph

The epigraph of a function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$, denoted by $\text{epi}(f)$, is the set of points in $\Omega \times \mathbb{R}$ given by

$$\text{epi}(f) = \{[x, \beta]^T : x \in \Omega, \beta \in \mathbb{R}, \beta \geq f(x)\}$$

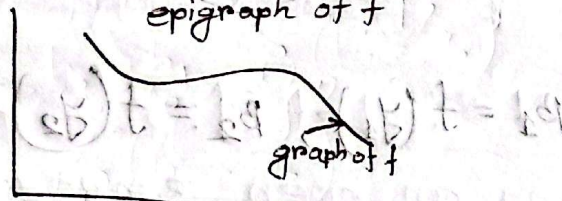


Fig: The graph and epigraph of a function $f: \mathbb{R} \rightarrow \mathbb{R}$.

Convex Set

A set $\Omega \subset \mathbb{R}^n$ is called convex if for every $x_1, x_2 \in \Omega$ and $\alpha \in (0, 1)$, $\alpha x_1 + (1-\alpha)x_2 \in \Omega$.

Convex Function

A function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$ is called convex on Ω if its epigraph is a convex set.

Theorem 21.1:- If a function $f: \Omega \rightarrow \mathbb{R}$, $\Omega \subset \mathbb{R}^n$ is convex on Ω , then Ω is a convex set.

Proof:- We prove the theorem by contraposition.

Suppose that Ω is not a convex set. Then there exist two points y_1 and y_2 such that for some $\alpha \in (0, 1)$,

$$z = \alpha y_1 + (1 - \alpha) y_2 \notin \Omega$$

Let

$$p_1 = f(y_1), \quad p_2 = f(y_2)$$

Then the pairs

$$\begin{bmatrix} y_1 \\ p_1 \end{bmatrix}, \begin{bmatrix} y_2 \\ p_2 \end{bmatrix}$$

belong to the graph of f and hence also the epigraph of f .

[Set $\rightarrow$ graph $\rightarrow$ epigraph]

$$\text{Let } w = \alpha \begin{bmatrix} y_1 \\ \beta_1 \end{bmatrix} + (1-\alpha) \begin{bmatrix} y_2 \\ \beta_2 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha y_1 + (1-\alpha) y_2 \\ \alpha \beta_1 + (1-\alpha) \beta_2 \end{bmatrix}$$

$$= \begin{bmatrix} \alpha y_1 + (1-\alpha) y_2 \\ \alpha \beta_1 + (1-\alpha) \beta_2 \end{bmatrix} \cdot \frac{1}{2} = b$$

But since $z \notin \Omega$, $w \notin \text{epi}(f)$. Therefore, $\text{epi}(f)$ is not convex and hence f is not a convex function.

→ **Thm 21.2**

Problem:- Let $f(x) = x_1 x_2$. Is f convex over

$$\Omega = \{x; x_1 \geq 0, x_2 \geq 0\}?$$

Solution:- Given that

$$f(x) = x_1 x_2$$

$$\text{and } \Omega = \{x; x_1 \geq 0, x_2 \geq 0\}$$

Let us consider $x = [1, 2]^T \in \Omega$ and $y = [2, 1]^T \in \Omega$

Then

$$2x + (1-2)y = 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (1-2) \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2-2 \\ 1+2 \end{bmatrix}$$

Then $f(2x + (1-\alpha)y) = (2-\alpha)(1+\alpha)$

and $2f(x) + (1-\alpha)f(y) = 2$

Let $\alpha = \frac{1}{2} \in (0, 1)$, then

$f(2x + (1-\alpha)y) = 2 + \frac{1}{2} - \frac{1}{4} = \frac{9}{4}$

Since

$f(2x + (1-\alpha)y) > 2f(x) + (1-\alpha)f(y)$

which shows that f is not convex over Ω .

Theorem 21.3:- Let $f: \Omega \rightarrow \mathbb{R}$, $f \in C^1$, be defined on an open convex set $\Omega \subset \mathbb{R}^n$. Then f is convex on Ω if and only if for all $x, y \in \Omega$

$f(y) \geq f(x) + Df(x)(y-x)$

Proof:- Suppose $f: \Omega \rightarrow \mathbb{R}$ is differentiable and convex. Then for any $y, x \in \Omega$ and $\alpha \in (0, 1)$ we have

$$f(\alpha y + (1-\alpha)x) \leq \alpha f(y) + (1-\alpha)f(x) \quad \text{or}$$

$$(1) \Rightarrow \alpha f(y) + (1-\alpha)f(x) \leq \alpha f(y) + f(x) - \alpha f(x)$$

$$(2) \Rightarrow \alpha f(y) + f(x) - \alpha f(x) \leq f(x) + \alpha(f(y) - f(x))$$

$$\Rightarrow f(x) + \alpha f(y-x) \leq f(x) + \alpha(f(y) - f(x))$$

$$\Rightarrow f(x + \alpha(y-x)) \leq f(x) + \alpha(f(y) - f(x))$$

$$(w-v)(1-\alpha) \Rightarrow f(x + \alpha(y-x)) - f(x) \leq \alpha(f(y) - f(x))$$

$$\Rightarrow \frac{f(x + \alpha(y-x)) - f(x)}{\alpha} \leq f(y) - f(x)$$

Now taking the limit as $\alpha \rightarrow 0$ and by the definition of the directional derivative of f at x in the direction $y-x$, we get

$$Df(x)(y-x) \leq f(y) - f(x)$$

$$\Rightarrow f(y) \geq f(x) + Df(x)(y-x)$$

Now conversely assume that Ω is convex, $f: \Omega \rightarrow \mathbb{R}$ is differentiable and for all $x, y \in \Omega$

$$f(y) \geq f(x) + Df(x)(y-x)$$

Let $u, v \in \Omega$ and $\alpha \in (0, 1)$. Because Ω is convex

$$w = \alpha u + (1-\alpha)v \in \Omega$$

We also have

$$(1) f(u) \geq f(w) + Df(w)(u-w) \quad (1)$$

$$(2) f(v) \geq f(w) + Df(w)(v-w) \quad (2)$$

Multiplying (1) by α and (2) by $(1-\alpha)$ and then adding, we get

$$\alpha f(u) + (1-\alpha)f(v) \geq f(w) + Df(w)(\alpha u + (1-\alpha)v - w)$$

$$\text{But } w = \alpha u + (1-\alpha)v \quad \text{so } \alpha u + (1-\alpha)v - w = 0$$

Hence

$$\alpha f(u) + (1-\alpha)f(v) \geq f(\alpha u + (1-\alpha)v)$$

Hence f is a convex function.

Theorem 21.5: - Let $f: \Omega \rightarrow \mathbb{R}$ be a convex function defined on a convex set $\Omega \subset \mathbb{R}^n$.

Then a point is a global minimizer of f over Ω (if and only if) it is a local minimizer of f .

Proof: - The first part of the theorem is obvious. We prove the second part by contraposition.

Suppose that x^* is not a global minimizer of f over Ω . Then for some $y \in \Omega$, we have $f(y) < f(x^*)$. By assumption, the function f is convex, and hence for all $\alpha \in (0, 1)$,

$$f(\alpha y + (1-\alpha)x^*) \leq \alpha f(y) + (1-\alpha)f(x^*)$$

Because $f(y) < f(x^*)$, we have

$$\alpha f(y) + (1-\alpha)f(x^*) = \alpha(f(y) - f(x^*)) + f(x^*) < f(x^*)$$

Thus for all $\alpha \in (0, 1)$

$$f(\alpha y + (1-\alpha)x^*) < f(x^*)$$

$$\begin{aligned} & \alpha f(y) + (1-\alpha)f(x^*) \\ &= \alpha f(y) + f(x^*) - \alpha f(x^*) \\ &= \alpha(f(y) - f(x^*)) + f(x^*) \\ & \text{Example: } f(y)=5, f(x^*)=7, \alpha=\frac{1}{2} \\ & \frac{1}{2}(5-7)+7 = -1+7 = 6 < 7 \end{aligned}$$

Hence there exist points that are arbitrarily close to x^* and have lower objective function value. For example, the sequence $\{y_n\}$ of points given by

$$y_n = \frac{1}{n}y + \left(1 - \frac{1}{n}\right)x^*$$

converges to x^* and $f(y_n) < f(x^*)$. Hence x^* is not a local minimizer, which completes the proof.

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convex > 0
 # concave < 0

Example 21.6: Determine whether the following functions are convex, concave or neither

1. $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -8x^2$

2. $f: \mathbb{R}^3 \rightarrow \mathbb{R}, f(x) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2 + x_1x_3 - 3x_1 - 2x_2 + 15$

3. $f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x) = 2x_1x_2 - x_1^2 - x_2^2$

Solution:

1. Given that

$f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = -8x^2$

Now $\nabla f(x) = -16x$

The Hessian matrix of f is

$F(x) = -16 < 0$, for all $x \in \mathbb{R}$.

Hence, f is concave over $\mathbb{R}$.

2. Given that

$f: \mathbb{R}^3 \rightarrow \mathbb{R}$

$f(x) = 4x_1^2 + 3x_2^2 + 5x_3^2 + 6x_1x_2 + x_1x_3 - 3x_1 - 2x_2 + 15$

Now $\nabla f(x) = \begin{bmatrix} 8x_1 + 6x_2 + x_3 - 3 \\ 6x_2 + 6x_1 - 2 \\ 10x_3 + x_1 \end{bmatrix}$

The Hessian matrix of f is

$$F(x) = \begin{bmatrix} 8 & 6 & 0 \\ 6 & 6 & 0 \\ 1 & 0 & 10 \end{bmatrix}$$

The leading principal minors of $F(x)$ are

$$\Delta_1 = 8 > 0$$

$$\Delta_2 = \det \begin{bmatrix} 8 & 6 \\ 6 & 6 \end{bmatrix} = 12 > 0$$

$$\Delta_3 = \det F(x) = 114 > 0$$

Hence, $F(x)$ is positive definite for all $x \in \mathbb{R}^3$.

Therefore, f is a convex over $\mathbb{R}$.

3. Given that

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}, f(x) = 2x_1x_2 - x_1^2 - x_2^2$$

$$\text{Now } \nabla f(x) = \begin{bmatrix} 2x_2 - 2x_1 \\ 2x_1 - 2x_2 \end{bmatrix}$$

The Hessian of f is

$$F(x) = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} = 0$$

$$\Delta_1 = -2$$

$$\Delta_2 = 0$$

which is negative semidefinite for all $x \in \mathbb{R}^2$.

Hence f is concave on $\mathbb{R}^2$.

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Quadratic Form to verify convexity

A quadratic form $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is a function

$$f(x) = x^T Q x$$

where Q is an $n \times n$ real matrix.

Proposition 21.1:- A quadratic form $f: \Omega \rightarrow \mathbb{R}$,

$\Omega \subset \mathbb{R}^n$, given by $f(x) = x^T Q x$, $Q \in \mathbb{R}^{n \times n}$, $Q = Q^T$ is convex on Ω if and only if for all $x, y \in \Omega$

$$(x-y)^T Q (x-y) \geq 0.$$

Solution:- We know that a function is convex if and only if for every $\lambda \in (0, 1)$ and every $x, y \in \mathbb{R}^n$, we have

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

$$\Rightarrow \lambda f(x) + (1-\lambda)f(y) - f(\lambda x + (1-\lambda)y) \geq 0$$

$$= \begin{bmatrix} \lambda x & (1-\lambda)y \end{bmatrix} \begin{bmatrix} Q & 0 \\ 0 & Q \end{bmatrix} \begin{bmatrix} \lambda x \\ (1-\lambda)y \end{bmatrix}$$

[Calculation (1)(2)(3)]

Substituting $f(x) = x^T Q x$ into the left-hand side of the above equation,

$$\begin{aligned} & \alpha x^T Q x + (1-\alpha) y^T Q y - (\alpha x + (1-\alpha) y)^T Q (\alpha x + (1-\alpha) y) \\ &= \alpha x^T Q x + y^T Q y - \alpha y^T Q y - \alpha^2 x^T Q x \\ & \quad - (2\alpha - 2\alpha^2) x^T Q y - (1 - 2\alpha + \alpha^2) y^T Q y \\ &= \alpha(1-\alpha) x^T Q x - 2\alpha(1-\alpha) x^T Q y + \alpha(1-\alpha) y^T Q y \\ &= \alpha(1-\alpha) (x-y)^T Q (x-y) \end{aligned}$$

Therefore, f is convex if and only if

$$\alpha(1-\alpha) (x-y)^T Q (x-y) \geq 0$$

$$\Rightarrow (x-y)^T Q (x-y) \geq 0$$

which proves the result.

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Calculation:

$$\begin{aligned}
& 2x^T \alpha x + (1-2)y^T \alpha y - (2x + (1-2)y)^T \alpha (2x + (1-2)y) \\
&= 2x^T \alpha x + y^T \alpha y - 2y^T \alpha y - [(2x^T + y^T - 2y^T) \alpha (2x + (1-2)y)] \\
&= 2x^T \alpha x + y^T \alpha y - 2y^T \alpha y - [2\tilde{x}^T \alpha x + 2\tilde{x}^T \alpha y - 2\tilde{x}^T \alpha y \\
&\quad + 2y^T \alpha x + y^T \alpha y - 2y^T \alpha y - 2\tilde{y}^T \alpha x \\
&\quad - 2y^T \alpha y + 2\tilde{y}^T \alpha y] \\
&= 2x^T \alpha x + y^T \alpha y - 2y^T \alpha y - [2\tilde{x}^T \alpha x + (2-2\tilde{y})x^T \alpha y \\
&\quad + (2-2\tilde{y})y^T \alpha x + (1-2\alpha + \tilde{y})y^T \alpha y] \\
&= (2-2\tilde{y})x^T \alpha x + (2-2\tilde{y})y^T \alpha y - (2\alpha - 2\tilde{y})x^T \alpha y \\
&= 2(1-\alpha)x^T \alpha x - 2\alpha(1-\alpha)x^T \alpha y + \alpha(1-\alpha)y^T \alpha y \\
&= 2(1-\alpha)(x^T \alpha x - 2x^T \alpha y + y^T \alpha y) \\
&= 2(1-\alpha)(x^T \alpha x - x^T \alpha y - \underline{x^T \alpha y} + y^T \alpha y) \\
&= 2(1-\alpha)(x^T \alpha x - x^T \alpha y - \underline{y^T \alpha x} + y^T \alpha y) \\
&= 2(1-\alpha)[x^T \alpha (x-y) - y^T \alpha (x-y)] \quad [x^T \alpha y = y^T \alpha x = \text{Same Result}] \\
&= 2(1-\alpha)(x-y)^T \alpha (x-y)
\end{aligned}$$